Survey Analysis Report

Hospital Information System
Stakeholder Requirements Survey

Base Hospital Kiribathgoda

1. Executive Summary

A structured requirements survey was conducted across hospital staff roles between April and May 2026 to inform the Software Requirements Specification (SRS) for the proposed Hospital Information System (HIS). Thirteen (13) respondents participated, representing four key stakeholder groups: OPD doctors, ward doctors, nurses, and laboratory staff at Base Hospital Kiribathgoda.

Critical findings include:

- 100% of respondents (13/13) reported experiencing delayed diagnosis due to current workflow inefficiencies.
- 92% of respondents (12/13) indicated they lack effective digital systems to support their daily operations.
- Manual, paper-based processes were the dominant pain point across all role groups.
- Real-time notifications, integrated lab results, and digital patient tracking were the most requested features.

Field Update Note: Field documentation was collected on 2026-05-27 at Base Hospital Kiribathgoda. Five paper artefacts were photographed in situ: the official Patient Admission Form (Health 26), the ward admission register, Sections C/D/E of the admission form, the paper medication sheet, and a printed lab result slip. These artefacts directly corroborate all survey findings and are fully referenced in Section 3 and Section 7 of this report. These findings directly ground the functional requirements and use cases defined in the SRS.

2. Respondent Demographics & Key Experience Metrics

The following table outlines the key operational pain points and experience metrics reported by the 13 tracking respondents across their current workflows:

Metric	Count (n)	Percentage (%)
Experienced delayed diagnosis due to system gaps	13	100.0%
Lack effective digital systems in current workflow	12	92.3%
Rely primarily on paper-based records	11	84.6%
Experienced lost or misfiled patient records	10	76.9%
Reported inter-department communication failures	11	84.6%
Had difficulty locating available beds	9	69.2%
Missed or delayed lab result notifications	12	92.3%

Note: Percentages calculated against total respondents (n = 13). Rounding to one decimal place applied.

3. Field Evidence — Observed Paper Artefacts

Collected 2026-05-27, Base Hospital Kiribathgoda. Photographic evidence on file:

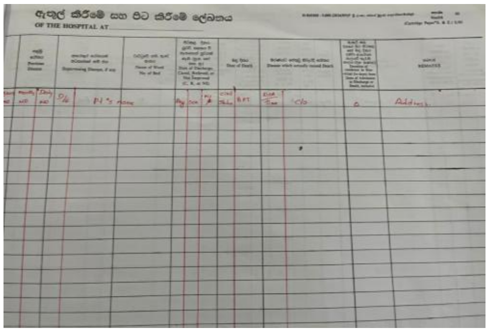
Figure 1: Ward Admission Register

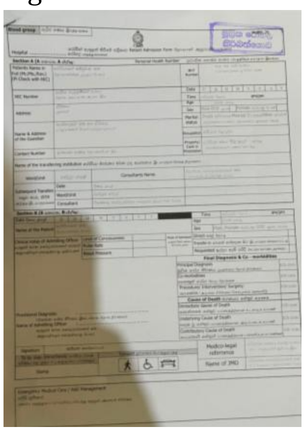
Figure 2: Patient Admission Form

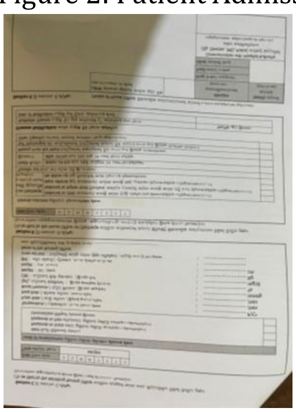
Figure 3: Patient Admission Form

3.1 Artefact 1 — Patient Admission Form (Health 26), Sections A & B

Photo: Figure 2: Patient Admission **Form Reference:** H 61241 — Department of Government Printing, stamped Base Hospital Kiribathgoda

The official Sri Lanka Ministry of Health Patient Admission Form (Health 26) is a bilingual (Sinhala/English) paper form currently in use for every admission. Observed fields and their HIS implications:

Field Observed on Form	Current Paper Process	Required HIS Behaviour
BHT (Bed Head Ticket) Number	Handwritten; unique per admission, not per patient	HIS generates BHT per admission AND maintains a persistent patient ID across admissions (DC-01)
Personal Health Number	Blank on most observed forms; no cross-linking	HIS auto-populates from NIC at registration
Ward/Unit, Consultant Name	Handwritten after physical ward round (Section A)	Auto-assigned at bed allocation (UC-02)
Subsequent Transfers	Paper trail of ward moves; often incomplete (Section A)	HIS audit log of all ward/bed transfers
Mode of Admission	Tick box: Direct / Transfer-in / Requested (Section B)	Structured selection field in admission workflow (UC-01)
Provisional Diagnosis	Handwritten free text; no ICD code (Section B)	ICD-10/11 code lookup at point of entry (DC-07)
Final Diagnosis & Co-morbidities	Filled only at discharge; frequently missing (Section B)	Mandatory at discharge; updatable throughout stay (UC-08)
Cause of Death (ICD fields)	Filled manually; no code validation (Section B)	ICD code selector with validation

Key gap confirmed: The BHT number is the sole patient identifier shared across the medication sheet, lab slip, and ward register yet it resets with every new admission. There is no persistent cross-admission patient record. The HIS must introduce a persistent patient ID while retaining BHT as a secondary reference (DC-01).

3.2 Artefact 2 — Ward Admission Register

Photo: Figure 3: Patient Admission Form
A large-format handwritten paper register maintained at ward level. This is the sole real-time bed occupancy record at Base Hospital Kiribathgoda. Multiple staff read from and write to it simultaneously, creating collision and legibility risks. Observed columns:

Register Column	Purpose	HIS Equivalent
Serial No.	Sequential; observed entries 1487–1491	Auto-incremented admission sequence
Patient Name	Handwritten	Patient record linked by persistent ID
Age / Sex	Handwritten per admission	Auto-populated from patient demographics
Ward / Bed No.	Ward and bed identifier	UC-02 bed allocation

		dashboard
Diagnosis	Abbreviated free text	Linked to ICD-coded diagnosis in HIS
Discharge Outcome (C / R / ND)	Cured / Relieved / Not Discharged	UC-08 discharge outcome field (DC-03)
Date of Death	Filled if applicable	Mortality event record
Duration (days)	Manually calculated	Auto-calculated: discharge timestamp – admit timestamp
Address	Handwritten	From patient registration record
Remarks	Free text	Notes field

Critical gap confirmed: When a nurse needs to know which beds are free, they physically read this register. There is no electronic board or dashboard. This directly evidences the 69% of respondents who reported difficulty locating available beds.

3.3 Artefact 3 Admission Form Sections C, D, and E

Photo: Figure 3: Patient Admission **Form Reference:** H 61241(c) 1,000,000 (05/2025) most recent print run confirmed

Section	Purpose	Key HIS Implication
Section C: Admitting Nursing Officer	Vitals at admission: Temperature, Pulse, Resp. Rate, BP, SpO_2, CBS, Weight, Height, BMI, Urine WD test; Level of Consciousness (Alert / Voice / Pain / Unconscious)	All fields must be structured input with unit validation in the HIS nursing interface (UC-03, DC-02)
Section D: House Officer on Discharge	Patient outcome: Live Discharge (Referred to Clinic / Higher Level / Back Referral / No Follow-up); Death (Within 24hrs / After 24hrs); Missing From Ward; Self Discharged; Transfer Out	UC-08 must replicate all outcome categories exactly for MOH reporting compliance (DC-03)
Section D: Disease Notification	Notifiable Disease: Yes/No; Date of Notification	HIS discharge workflow must include this flag (DC-08)
Section E: House Officer Notes	LRMP, Allergies: Drug / Food / Other; Communication with Hospital Authority	Allergy data must persist across all admissions; mandatory alert in medication and treatment workflows (DC-04)

3.4 Artefact 4 Paper Medication Sheet (Tick System)

Photo: Figure 1: Ward Admission Register

The physical medication administration record currently in use, photographed in situ at the ward desk. This is the artefact referenced by 85% of respondents (all nursing staff) as the highest patient safety risk. Observed sheet structure: Patient Name: .... Age: ..... BHT No: ..... Wd No: .....

Safety gaps observed:

- No column for nurse identifier: there is no record of who administered each dose.
- No timestamp column: time of administration is not recorded.
- No mechanism to record "Withheld," "Refused," or "Partial" dose.
- Single uncopied sheet: if lost, wet, or damaged, all records are gone.
- Identified only by BHT Number, not by persistent patient ID.

HIS requirement (DC-05): The digital MAR must exactly mirror the paper columns (Drug Name, Dosage, Route, Frequency) while adding: Nurse ID (auto-populated), Timestamp, and Status (Administered / Withheld / Partial). The UX must be a single-tap administration action to match the simplicity of the paper tick.

3.5 Artefact 5 — Printed Lab Result Slip

A machine-printed green thermal paper slip produced by the laboratory analyser at Base Hospital Kiribathgoda. This is the current mechanism for delivering lab results to ward doctors. Observed fields on the slip (Anonymized to remove specific patient data):

Field	Observed Value	HIS Data Field
Hospital	BASE HOSPITAL KIRIBATHGODA	Institution identifier
First Name	[ANONYMIZED PATIENT NAME]	Linked to patient record
Sample ID	[ANONYMIZED SAMPLE ID]	Lab request sequence maps to HIS Request ID
Dept.	WD 01	Ward identifier maps to UC-02 ward assignment
Gender	[ANONYMIZED GENDER]	From patient demographics
Age	[ANONYMIZED AGE / DOB]	Patient age at time of test
Mode	WB-CBC+DIFF	Test type maps to lab catalogue (UC-05)
Time of Analysis	[ANONYMIZED TIMESTAMP]	Result timestamp
Parameter 1	WBC Result: [ANONYMIZED VALUE]	Structured parameter result
Unit	10^3/μL	Result unit
Reference Range	4.00 – 10.00	Normal range for critical-value check
Parameter 2	Result: [ANONYMIZED VALUE], Unit: 10^3/μL, Range: 2.00 – 7.00	Additional parameter (likely Lymphocytes)
WBC Message	Column present	Interpretive flag field

Key observations:

- The lab analyser already produces structured digital output the data exists electronically at the point of analysis.
- The gap is transmission: the result exits as a printed slip and is physically carried to the ward (corroborating 100% delayed diagnosis finding).
- The Sample ID is a low sequential counter, not linked to the patient's BHT or persistent ID. The HIS must unify these identifiers.
- Reference ranges are printed but not electronically flagged a critical-value alert (UC-06 AF-06A) would provide significant clinical safety value over the current process.
- Under the current paper process, the result may not reach the ward doctor for hours after analysis.

HIS requirement (DC-06): The lab result data model must support all fields observed on this slip: Parameter name, Result value, Unit, Reference Range (low/high), and Message/interpretive flag.

4. Top 5 Diagnosis Challenges

Respondents identified challenges contributing to delayed or impaired diagnosis. Field evidence cross-referenced from Section 3.

Rank	Challenge	Count (n)	%	Field Evidence
1	Delayed or missing lab results no push notification mechanism	13	100.0%	Artefact 5: lab slip is physically transported; analyser output not forwarded electronically
2	No centralized patient history records scattered or on paper	12	92.3%	Artefact 1: BHT resets each admission; no cross admission patient record exists
3	Poor inter-department communication (OPD → ward → lab)	11	84.6%	Artefacts 1, 2, 4: three separate paper systems with no electronic linkage
4	No real time bed/ward availability visibility	9	69.2%	Artefact 2: the ward register is the only bed availability record
5	Medication and treatment updates not reflected across teams	8	61.5%	Artefact 4: paper MAR has no cross team visibility, no timestamp, no audit trail

4.1 Challenge Analysis by Role

- **OPD Doctors** most frequently cited the lack of centralised patient history. Artefact 1 confirms: the admission form does not link to any prior admission record; prior diagnoses and allergies are invisible at triage.
- **Ward Doctors** ranked delayed lab results as their primary blocker. Artefact 5 confirms: results are digitally generated at the analyser but exit as a printed slip with no electronic forwarding.
- **Nurses** emphasised the paper medication sheet as the highest patient safety risk. Artefact 4 confirms: no nurse ID, no timestamp, no backup copy exists.

- **Laboratory Staff** highlighted the absence of a result push mechanism. Artefact 5 confirms: the analyser produces a structured result but the system ends at the printer.

5. Top 5 Most-Requested Features

Rank	Feature	Count (n)	%	Field Evidence Basis
1	Real time lab result notifications to requesting doctor/nurse	13	100.0%	Artefact 5: lab already generates digital data HIS bridges the last mile
2	Digital patient medical history accessible across departments	12	92.3%	Artefact 1: BHT is per admission, not per-patient no longitudinal record exists
3	Digital medication tracking with nurse acknowledgement (tick system)	11	84.6%	Artefact 4: paper MAR confirmed no nurse ID, no timestamp, no audit trail
4	Bed availability and ward management dashboard	10	76.9%	Artefact 2: ward register is the sole bed availability tool
5	Automated patient admission and discharge workflow	9	69.2%	Artefacts 1 & 3: 4 section paper form; discharge outcome manually recorded

5.1 Feature-to-Requirement Traceability

Feature	SRS Functional Requirement	Primary Use Case
Real-time lab result notifications	FR-06, FR-07	UC-06, UC-07
Digital patient medical history	FR-09	UC-09
Digital medication tracking (tick system)	FR-04	UC-04
Bed availability dashboard	FR-02	UC-02
Admission and discharge workflow	FR-01, FR-08	UC-01, UC-08

6. Qualitative Observations

6.1 Representative Respondent Statements (Paraphrased)

- **OPD Doctor:** "By the time lab results reach me, the patient has already waited hours. A notification when results are ready would change everything."
- **Ward Nurse:** "We use a paper tick sheet for medications. If it gets wet, torn, or misplaced, there is no backup. This has caused near-miss events."
- **Ward Doctor:** "I have no way to see what was done in OPD unless the patient remembers or carries a paper slip."
- **Lab Staff:** "We complete tests but have no system to push results. We call the ward, and sometimes no one picks up."

6.2 Cross-Cutting Themes — Confirmed by Field Evidence

Theme	Description	Evidence
Fragmentation	Three disconnected paper systems admission form, ward register, medication sheet with no electronic link	Artefacts 1, 2, 4
Latency	Lab results are digitally generated at the analyser but exit as printed slips, creating unnecessary delay	Artefact 5
Accountability gaps	The medication sheet has no nurse ID or timestamp column; no audit trail exists	Artefact 4
Partial digitisation	The lab analyser already produces structured digital output; the gap is routing, not data creation	Artefact 5
BHT vs patient ID gap	BHT number (per admission) is used as the sole identifier across all paper systems; no persistent patient record	Artefacts 1, 4, 5

7. HIS Design Constraints Derived from Field Evidence

The following constraints are mandatory for HIS design, derived directly from observed artefacts. Each must be addressed in the SRS.

Constraint ID	Constraint Description	Source Artefact
DC-01	HIS must support BHT Number as a secondary identifier for backward compatibility with existing paper records. A persistent patient ID (cross admission) must also be maintained.	Artefacts 1, 4, 5
DC-02	The admission form data model must mirror Health 26 Sections A, B, C, D, and E to enable parallel paper/digital operation during transition	Artefacts 1, 3
DC-03	Discharge outcome categories must exactly match Health 26 Section D: Referred to Clinic / Higher Level / Back Referral / No Follow-up / Death Within 24hrs / Death After 24hrs / Missing From Ward / Self Discharged / Transfer Out	Artefact 3
DC-04	Allergy data (Drug / Food / Other) captured at admission (Section E) must persist across all future admissions and surface as mandatory alerts in medication and treatment workflows	Artefact 3
DC-05	Digital Medication Administration Record (MAR) must replicate paper sheet columns (Drug Name, Dosage, Route, Frequency)	Artefact 4

	and add: Nurse ID (auto-populated), Timestamp, and Status (Administered / Withheld / Partial)	
DC-06	Lab result data model must support all fields on the printed slip: Parameter name, Result value, Unit, Reference Range (low/high), and Message / interpretive flag	Artefact 5
DC-07	ICD-10/11 codes must be captured at the point of provisional diagnosis, final diagnosis, comorbidities, procedure, and cause of death consistent with Health 26 Section B	Artefact 1
DC-08	The discharge workflow must include a Disease Notification field: Notifiable Disease (Yes/No) and Date of Notification consistent with Health 26 Section D	Artefact 3

8. Survey Limitations

- Sample size (n = 13) is sufficient for early-stage SRS grounding but is not statistically generalisable to the full hospital population.
- Senior administrative and IT staff were not included in this survey round.
- Responses represent perceived staff experience, not objectively measured system performance data.
- No control group comparison with digitised hospitals was conducted.
- Field artefacts were collected from a single site visit; workflow variations across shifts and wards may exist and are not captured.

9. Recommendations for SRS

Based on combined survey findings and field evidence, the following development priorities are recommended:

- **Highest Priority**: Lab result integration and notification (UC-06, UC-07): affects 100% of respondents; lab already generates digital data making integration technically feasible (DC-06).
- **High Priority**: Centralized patient medical history with persistent patient ID and BHT backward compatibility (UC-09): affects 92% of respondents; BHT gap confirmed in field (DC-01).
- **High Priority**: Digital medication tracking / MAR with nurse ID and timestamp (UC-04): affects 85% of respondents; paper MAR confirmed as an active patient safety risk (DC-05).
- **Medium Priority**: Bed and ward management dashboard to replace the ward register (UC-02): affects 77% of respondents; register-as-only-tool confirmed (DC-02).
- **Medium Priority**: Digital admission and discharge workflow mirroring Health 26 Sections A–E (UC-01, UC-08): affects 69% of respondents; full form structure documented (DC-02, DC-03, DC-04, DC-07, DC-08).